PHYSICS 6031, ELECTRODYNAMICS I
SPRING 2023

Lecturer: Daniel Phillips
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Class times & location: 11:00 am-12:20 pm, Tuesday & Thursday in Gordy 209 and 10:45-11:40 am Friday in Gordy 201
Office hours: 3-4 pm Monday & 2-3 pm Tuesday, or by appointment
Textbook (required): *Classical Electrodynamics*, by J.D. Jackson, 3rd edition (Wiley, 1998).
Course website: http://www.phy.ohio.edu/~phillips/Electrodynamics.html
Class number: 8136 & 8148

Supplementary reading:
Introduction to Electrodynamics, by D.J. Griffiths (Prentice-Hall, 1981)
Classical Electricity and Magnetism, by W.K.H. Panofsky and M. Phillips (Addison-Wesley, 1955)

This semester we will discuss the classical theory of electromagnetism. I find it remarkable that work which started thousands of years ago, with observations of static electricity and the magnetic properties of lodestone, led us to think about lightning and currents, and then we (meaning Faraday, Maxwell, and others) realized all these phenomena are manifestations of the same underlying force: electromagnetism. The mathematics that unifies these different phenomena is also beautiful: certainly the greatest flowering of classical field theory up until 1900. And, as a bonus that mathematics describes light is a wave-like disturbance of the electromagnetic field.

Approximate schedule:
The following schedule is *tentative*. I will adjust as I see how we progress through the semester.

Week beginning:	Topic and Jackson Chapter
January 17	Motivation, Chapter I; Review of Undergraduate Electrostatics, Chapter 1
January 23	Poisson and Laplace Equations, Chapter 1; Method of Images, Chapter 2
January 30	Application of Math Methods to Laplace's equation, Chapters 2 and 3
February 6	Multipoles, Chapter 4

Week beginning:	Topic and Jackson Chapter
February 13	Dielectrics, Chapter 4
February 20	Midterm 1 + Undergraduate Magnetostatics, Chapter 5
February 27	Magnetostatics; Magnetism in Media, Chapter 5
March 6	Faraday's Law; Energy stored in the Electromagnetic field, Chapter 6
March 13	SPRING BREAK
March 20	Maxwell's equations, Chapter 6
March 27	Electromagnetic waves, Chapter 7 + Midterm 2
April 3	Dispersive media & dispersion relations*, Chapter 7; Undergraduate Special Relativity, Chapter 11
April 10	Special Relativity in terms of four vectors, Chapter 11;
April 17	Dynamics of relativistic particles in electromagnetic fields, Chapter 12; Radiation by moving charges, Chapter 14;
April 24	Radiation by antennas*, Chapter 9; Wave guides*, Chapter 8

This is a *tentative* schedule. Topics with a * may be sacrificed if we need to spend more time on other things.

Learning outcomes:

- Apply mathematical methods to problems in electrodynamics.
- State the principles of special relativity and apply them to particle motion.
- Compute the radiation field of a charged particle.
- Calculate the response of charged particles and polarizable matter to electromagnetic fields.
- Be able to derive vector fields for given sources and boundary conditions.

Weekly structure:

I will send out the reading you are expected to do before class and then we will discuss it in class. I may include some questions to think about when I send out the reading. Please also come to class with questions for me to answer about things that puzzled you in the reading. During class I will spend less than half the time "lecturing". We'll spend significant time in small groups, doing examples and working through homework-like problems. We will continue in the same mode in Friday's class, except there we will also discuss homework problems that caused difficulty.

Attendance and participation in class:

This term it is especially critical that you not attend class in-person when ill. If you do test positive or need to isolate or quarantine this semester, after you have taken care of yourself and followed all the steps in the OHIO COVID-19 Protocol, please email me so that we can develop a plan for you to receive necessary course content. COVID-related illness, quarantine, isolation and remain-in-room directives are "legitimate university absences" (cf. also discussion of late work below). If you feel that your class performance is being impacted by COVID-19, please talk with and/or email me. The University has information about resources available to help with quarantine and isolation here.

I do not take attendance. But if you miss a class because you're ill, or for any other reason, please do contact me or a classmate to make sure you know what was presented **and also know the content of any announcements that were made**.

Participation expectations

- You are expected to have done the reading (and thought about some questions to ask!) before coming to class.
- You should take notes during the classes where we discuss material. Taking notes is a key part of your growth as scholars: learning how to extract the main points from a presentation and write them down in real time is a skill you should develop through practice. And you should plan to take notes on paper. Studies show that taking notes on a laptop or other electronic device does not aid retention as fully as taking notes on paper does.
- During class please switch you cell-phone to vibrate and put it away. I want you to be fully engaged in the class, ready to ask—and answer—questions.
- You are expected to turn in homework on time. Late or make-up work is accepted only in case of illness, death in the immediate family, religious observance, jury duty, and involvement in University-sponsored activities (departmental trip, music or debate activity, ROTC function, or athletic competition) and service or training for military reserves, including reasonable travel time to the training location. These reasons also constitute legitimate reasons for missing a class. You may document reasons for your absence as explained in the "Academic Policies and Procedures" portion of the Graduate Catalog.
- In case you have to quarantine and attend class remotely should make sure you have a laptop or tablet with a functioning camera and microphone, and that Zoom is installed on your device.

10% of your grade will be based on participation in and preparation for the class sessions.

Homework:

The homework assignments are an integral component of the work for this class. To learn this subject you need to practice the mathematical description of electromagnetism and understand its connection to physical phenomena. You will do that by solving problems. This also means that understanding how to solve the homework problems is important preparation for the exams. We will spend time discussing some of the homework problems in class, but if you are still confused

after that discussion, or you're confused about a problem we don't cover there, I strongly encourage you to discuss it with me or a classmate. In particular, you should avail yourself of the office hours. I do not expect you to solve all these problems on your own, and am happy to try and get you headed in the right direction if you are having difficulties.

If you are struggling to complete your homework on time please let me know in advance of the deadline, and we will try to work something out. Homework assignments will generally be due first thing (let's say by 10 am) on Wednesday. The problem sheets will be handed out one week before they are due. The graded assignments will be returned as soon as possible. We will discuss homework problems that people found challenging during Friday's class meeting. During that discussion I may ask a student to present their solution to the rest of the class.

In grading your written homework solutions I will be looking to see if you understand how to solve the problems. Therefore partial credit will be given for incomplete solutions, and, conversely, the correct answer without adequate explanation will actually yield very little credit. All steps used in reaching the solution must be properly explained and justified. The solution should be able to be read as a coherent discussion in English of the problem. I.e., explanatory sentences should be inserted into mathematical reasoning. You should also, where appropriate, draw a diagram to illustrate the physical situation you are describing in your problem solution. Homework assignments count for 20% of the final grade.

Collaborating on homework with other students:
Plagiarism (i.e. copying each other's solutions or lab reports) will not be tolerated. However, I think it can be a good idea for you to discuss together how to solve problems. Operationally, this means that you are welcome to talk to each other about your assignments, but you should each go off and generate your own solutions to be handed in. See also discussion of plagiarism below, on page 6 of the syllabus.

Exams:
There will be two midterm exams and a final exam. The midterms will each be worth 20%. The final exam will be worth 30%. The exams will be closed book and will each last two hours. All exams will take place in Gordy 209, i.e., in our regular classroom. Midterm 1 will take place on Tuesday, February 21 at 10:15 am.
Midterm 2 will take place on Thursday, March 30 at 10:15 am.
The final exam will take place Tuesday, May 2 at 11 am.

At the exams students are expected to supply pens or pencils, scratch paper, and a calculator. These exams will be "closed book": no books, notes, or formulas stored in electronic or written form may be consulted during them.

Summary of grading:

Weekly homework:	20%
Midterm 1 (Week 6):	20%
Midterm 2 (Week 11):	20%

Final exam (Week 16): 30%
Class preparation & participation: 10%

The value of inclusion and equity
The best environment in which to learn physics is one that's inclusive and equitable. The students, staff, and faculty of the Department of Physics and Astronomy are committed to professional interactions, with respect and consideration of the rich and diverse backgrounds of all its members. We expect everyone in the department, including students taking classes, to encourage & support a culture that—no matter what the activity is—-includes and treats equally all social identities. We also uphold all Ohio University's diversity policies. For more information about our culture of inclusion and how to report issues and concerns, consult the Inclusion and Equity website of the Physics and Astronomy Department.

Accommodations for students with special needs:
If you suspect you may need an accommodation based on the impact of a disability please contact me privately to discuss your specific needs. In general, I need written documentation from the Office of Student Accessibility Services in order to determine the appropriate accommodation. If you have not yet registered as a student with a disability then you should contact that office. If your learning needs are such that you wish to receive an exception from my electronic-device policy (see above) then please talk to me and we will determine a suitable arrangement.

Academic misconduct:
"Academic integrity and honesty are basic values of Ohio University. Students are expected to follow standards of academic integrity and honesty. Academic misconduct is a violation of the Ohio University Student Code of Conduct subject to a maximum sanction of disciplinary suspension or expulsion as well as a grade penalty in the course". Academic misconduct is defined as "dishonesty or deception in fulfilling academic requirements. It includes, but is not limited to: cheating, plagiarism, un-permitted collaboration, … fabrication (e.g., use of invented information or falsification of research or other findings), using advantages not approved by the instructor (e.g., unauthorized review of a copy of an exam ahead of time), knowingly permitting another student to plagiarize or cheat from one's work, or submitting the same assignment in different courses without [instructor] consent".

"Cheating is defined as any attempt by a student to answer questions on a test, quiz, or assignment by means other than his or her own knowledge. Examples: using the textbook or other materials, such as a notebook, not authorized for use during an examination; using technology (i.e. cell phones, laptop computers, social media, text messages, etc.) to aid in the completion of work when not permitted to do so; observing the work of another student or allowing another student to plagiarize, copy, or observe your work; using unauthorized material during a test, notes, formula lists, notes written on clothing, etc.; taking a quiz, exam, or similar evaluation in the place of another person; providing or requesting assistance from another person in a manner prohibited by the instructor; using a laboratory, computer, or calculator improperly

or without authorization; changing material on a graded exam and then requesting a regarding of the exam; acquiring unauthorized knowledge of an examination or any part of an examination."

"Plagiarism is defined as the presentation of the ideas or the writing of someone else as one's own. Examples: reproducing another person's work, whether published or unpublished (this also includes using materials from companies that sell research papers); submitting as your own any academic exercise (written work, computer printout, sculpture) prepared totally or in part by another; allowing another person to substantially alter or revise your work and submitting it as your own; using another's written ideas or words without properly acknowledging the source. If a student uses the words of someone else, he or she must put quotation marks around the passage and add indication of its origin, such as a footnote; simply changing a word or two while leaving the organization and content substantially intact and failing to cite the source is plagiarism…."

See here for original source and further discussion.

In practice, this means you **cannot** copy solutions to problems from those with whom you are taking the class, or from others who have taken the class before you. Plagiarism can also occur if you **use a solution or approach that you read in a book or on a website and fail to properly cite the source**. If you base your answer to a problem substantially on something you have read **you must cite that source**.

I *do* think it is good for you to discuss together how to solve the problems. Operationally, this means you are welcome to talk together about the homework assignments, but then you should each go off and generate the solutions you will hand in on your own. In general, if you are unsure about a question of plagiarism or cheating, you are obligated to consult me on it before submitting your work.

Plagiarism and cheating will not be tolerated. If I catch you engaging in these I will not warn you before penalizing you. You may appeal any penalties I impose through the grade appeal process. I may also report you to the Office of Community Standards and Student Responsibility, which could choose to impose additional sanctions.

The lectures, classroom activities, and all materials associated with this class and developed by the instructor are copyrighted in the name of Daniel Phillips on January 12, 2023.